

Using the recurrent neural networks to predict the results of classical molecular dynamics modeling of crystalline solids

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ABSTRACT

In this article, on an example of modeling the equilibrium state of crystalline copper, the possibility of predicting the atomic configuration at the next time step using data on atomic configurations at several previous time steps by a bidirectional recurrent neural network trained on classical molecular dynamics data is shown. A network with a minimum number of hidden layers (the number of layers is three) is used, which corresponds to the required number of nodes for calculating velocity and acceleration when integrating particle motion equations using the finite difference method. The figure of merit of the simulation is based on a comparison of the dynamic structural factor of the system, calculated from the initial molecular dynamic trajectories, and the one calculated from the molecular trajectories predicted by the recurrent neural network. The graphs of the radial distribution function, as well as the histograms of the distribution of the first-order differences of coordinates, constructed from the initial data on the arrangement of atoms and from the data predicted by the neural network, show a good match. Both the graphs of the radial distribution function and the histograms of the distribution of the first-order coordinate differences do not undergo significant changes during the simulation of the equilibrium state (180 ps), which indicates an adequate representation of statistical patterns in molecular trajectories predicted by a recurrent neural network.

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I. INTRODUCTION

Molecular dynamics (MD) modeling is widely used to study the elastic and durability properties of materials, phase transitions, heat and mass transfer, wave propagation, relaxation kinetics to equilibrium state of systems after an external action, and other processes and phenomena in various fields of physics and chemistry (see, for example, Refs. 1–3). In modern studies of physical phenomena using numerical modeling, calculations are always performed at the limit of available computing power. Currently, systems containing up to 10^9 – 10^{10} particles can be modeled using classical MD (CMD), which, of course, is significantly less than

the number of particles that statistical physics usually works with, but it allows us to obtain results that correlate with experiments. Increasing the calculation speed is one of the conditions for the rise the maximum number of particles in the model system available for calculations and, respectively, to develop the practice of applying the molecular dynamics. Classical MD is based on the numerical solution of Newton's equations, which means that for a correct calculation, it is necessary to have the correct values of the forces acting on a particle. In fact, the high computational cost in studying of large systems and the issue of choosing an adequate force field (also called interatomic potential) are the main problems of the

classical molecular dynamics method, and in the last decade artificial neural networks have been used to solve these problems.

An interatomic potential can be approximated by a neural network that was trained on data obtained from results of a quantum chemical calculation of forces acting on atoms. Such quantum chemical calculations, used for an accumulation the learning sample, are carried out for fragments of the system under study or even the entire system (see, for example, Refs. 4–15). The Car–Parrinello¹⁶ or Born–Oppenheimer¹⁷ *ab initio* molecular dynamics (AIMD) data can be selected as the initial data for training. Fully connected,^{12,15} convolutional,⁴ graph^{4,13} networks, and their combinations are utilized to represent interatomic potentials (see, for example, Ref. 6). There are universal neural network potentials (for example, Refs. 5, 7, 8, and 13) which are used in modeling organic substances and, vice versa, potentials, based on neural network training methods and their architectures, specifically selected for a certain class of materials.^{9,10} The issues of representation by a neural network the multi-well potentials which use to modeling phenomena such as phase transitions, dynamic instabilities, and multimodal behavior are also being investigated.¹⁸

On the other hand, the classical MD (CMD) method is based on solving the equations of motion for each particle in a given field. These equations are second-order differential equations, therefore, to solve them (i.e., to obtain the position and velocities of all particles in the next time step), it is necessary to know the position of the particles in at least two previous steps. The set of frames of the particles' coordinates in consecutive time moments can be considered as time series so the techniques utilized in the field of machine learning can be used to analyze it. Recurrent neural networks (RNNs), which contain feedback loops and allow one to memorize the information,¹⁹ can be used to predict the next value of a time series. Theoretically, training these networks on the sequence of MD frames can make it possible to predict atomic configurations in the future. The use of recurrent neural networks is justified if the training and inference of the trained model is performed much faster than the actual MD calculation of the considered state is carried out. Currently, various research groups are investigating the use of recurrent neural networks of various types as a tool to accelerate and refine molecular dynamics.

Reference 20 is one of the first papers on the application of recurrent neural networks in the field of molecular dynamics. The authors of Ref. 20 use sequential frames of all atoms' positions in the unit cell of the $\text{Ir}_x\text{M}_{1-x}\text{O}_2$ sample (M is a metal atom) obtained from AIMD trajectories as dataset for neural network learning. At the network output, in addition to the atomic coordinates, fluctuations in the potential energy profile were calculated in Ref. 20, which were then compared with the data from AIMD calculations. The length of the sequence of frames of atomic positions in the packet, submitted to the input of the RNN layers, corresponds to the number of steps on the basis of which the new value was calculated. It ranged from 50 to 200 steps. It is noted in Ref. 20 that the main goal of AIMD modeling is to create as many statistically independent configurations of the system under study as possible at an acceptable computational cost. To achieve this goal, one should model a large number of states of the system under study and set sufficiently long intervals between configurations that are used in thermodynamic analysis to ensure that they are not statistically

correlated. In Ref. 20, it was established that RNN-based algorithms can generate trajectories with the same accuracy as in AIMD modeling, but in much less computational time. The approach presented in Ref. 20 considers interactions between atoms as temporal correlations; thus, the task of predicting the trajectories of atoms is reduced to the task of predicting multivariate time series. The most important result of Ref. 20 is the proof of the ability of a trained neural network to predict uncorrelating configurations that can be directly used to calculate thermodynamic properties.

In Ref. 21, a model is proposed for predicting the course of physical processes in the future based on the use of one of the types of RNN—a long short-term memory neural network (LSTM), trained on a small amount of data of classical MD modeling. This model extracts the latent vectors from the atomic coordinates of MD simulations using a convolutional (graph) neural network described in Ref. 22. The model utilizes LSTM to learn temporal trends in latent vectors and make one-step-ahead predictions of physical properties through fully connected layers. The model was tested on the simulation of the nickel solid–liquid system, and it showed the possibility of an accurate prediction of changes in potential energy over time during the solidification and melting processes, despite the fact that the MD data used in network training did not include frames corresponding to the phase transition. Recursive use of predicted values enabled long-term prediction from just the first 20 snapshots of the MD simulation. This prediction takes three orders less time than time needed for direct MD simulation.

Reference 23 discusses the use of data on the position of a particle in previous time steps to predict with RNN the position after a much longer time than the time step. The model has been tested to predict one-dimensional motion and the motion of several particles (up to 16) in a force field corresponding to the Lennard–Jones potential.

In Ref. 24, it was shown how molecular dynamic trajectories can be converted to the format of character sequences of a certain language. For these sequences, the methods to predict the next character with LSTM are developed. Using the inverse transformation of the predicted symbols to molecular trajectories, the authors of Ref. 24 obtain a statistically correct description of the system in the future.

In Refs. 25–32, recurrent neural networks in molecular dynamics are used to solve particular problems.

In Ref. 25, devoted to the AIMD simulation of liquid water, a RNN model for classifying ultrafast hydrogen bonding processes based on the data from MD trajectories was developed.

Reference 26 notes that when using RNN to represent the trajectories of molecular dynamics of large systems, rare events associated with “slow” motion may be unaccounted for in the model during training against the background of frequent events associated with “fast” motion.

In Ref. 27, using various types of bidirectional neural networks, the problem of interpolating the molecular trajectories of particles of a sample in contact with a thermostat is investigated. The results of the interpolation of the molecular trajectories are used in Ref. 27 to calculate the vibrational spectra, to refine the shape of free energy surfaces, and to increase detailization of the MD trajectories.

In Ref. 28, modeling of $\text{CdSe}_x\text{Te}_{1-x}$ solar cells with MD method is considered, and RNN of various types are used to speed up calculations.

TABLE I. Comparison of some recurrent neural networks used in molecular dynamics.

Reference	Architecture	Data source for training	Sequence length	The number of atoms in the system under study	The system under study	Model time	Relative error	Computational speedup
20	LSTM, GRU	AIMD (VASP)	250	36	$\text{Ir}_x\text{M}_{1-x}\text{O}_2$	5 ps	1% coordinates, 5% energy	
21	LSTM	CMD (LAMMPS)	20	20 000	Ni (solid-liquid system)	900 ps		700
23	LSTM	CMD	5	16	Particles under influence of Lennard-Jones potential	4000 Verlet timesteps	>1%-coordinates	32 000

In Ref. 29, the LSTM network is used to represent the molecular dynamics of the G-protein coupled receptor.

In Ref. 30, various types of RNN are used to predict molecular trajectories in modeling macromolecular polymer-lipid aggregate solvated in ethyl acetate at ambient conditions. It is shown that none of the neural networks used can achieve the required accuracy. A method is proposed for the experimental selection of an ensemble of neural networks trained on a set of time series, taking into account the previously identified time patterns of these series. A similar idea was suggested in Ref. 31, which consists in selecting a subset of the predicted trajectories generated on the basis of trained LSTM models that satisfy the specified thermodynamic constraints.

In Ref. 32, LSTM networks are used to predict the course of the oxidation process of uranium alloys as part of a complex model.

Thus, recurrent neural networks are widely used in research devoted to modeling various statistical distributions of particles, including as a means for predicting the configuration of particles at the next time steps of molecular dynamics. Table I shows the data of some of the above-mentioned works concerning the architecture, type, and size of the studied systems; the maximum model time achieved; the accuracy of calculations; and the acceleration of calculations compared with the direct implementation of molecular dynamics.

II. PROBLEM STATEMENT

The main idea of this paper proceeds from the following considerations. In the simplest case of numerical solution of a differential equation derived from Newton's laws, when tracing the evolution of a particle's coordinate to determine its next position, we need to know the current position of the particle, its velocity and acceleration at a given point, which is equivalent to knowing the force at the current point. The Verlet integration scheme used in classical molecular dynamics, instead of the requirement to know the velocity value, requires knowledge of two positions, at the current and previous time steps, and the acceleration of the particle. Most methods of applying neural networks in classical molecular dynamics are reduced to determining forces based on

the current molecular configuration. In this paper, an attempt is made to explore the possibility of using a recurrent neural network that will predict the next configuration of atoms only from several previous ones. In contrast to Ref. 20, a network with a minimum physically determined number of layers, namely, a three-layer network, will be used. Three consecutive positions are minimally necessary to obtain a prediction, precisely so that the network can "calculate" particles' accelerations during simulation.

III. METHODOLOGY

Figure 1 shows the diagram of the neural network. The network shown in Fig. 1 is a bidirectional recurrent neural network that accepts as input three consecutive configurations of the simulated system $x(t)$, $x(t-1)$, $x(t-2)$ at time points t , $t-\Delta t$, and $t-2\Delta t$, respectively, and returns the configuration of the system $x(t+1)$ at time $t+\Delta t$. To simplify the representation of the configuration of the system's state, crystalline bodies were considered, since it is convenient to take sets of coordinates of particle deviations from equilibrium positions (at the nodes of the crystal lattice) at the appropriate time as the configuration of the system under study

$$x(t) = \{\Delta x_1, \Delta y_1, \Delta z_1, \Delta x_2, \Delta y_2, \Delta z_2, \dots, \Delta x_N, \Delta y_N, \Delta z_N\}, \quad (1)$$

where N is the total number of particles in the system under consideration. This choice of the system's state representation is opportune for at least two reasons:

- (1) All particle deviations from equilibrium positions are of the same order of magnitude.
- (2) The symmetry of the spatial arrangement of atoms is automatically taken into account.

Copper with a face-centered cubic lattice was considered as a model system (Fig. 2). In Fig. 2(a), the atoms' positions are presented, and in Fig. 2(b), an arrangement of the points of high symmetry in the reverse space for a face-centered cubic lattice³³ is presented.

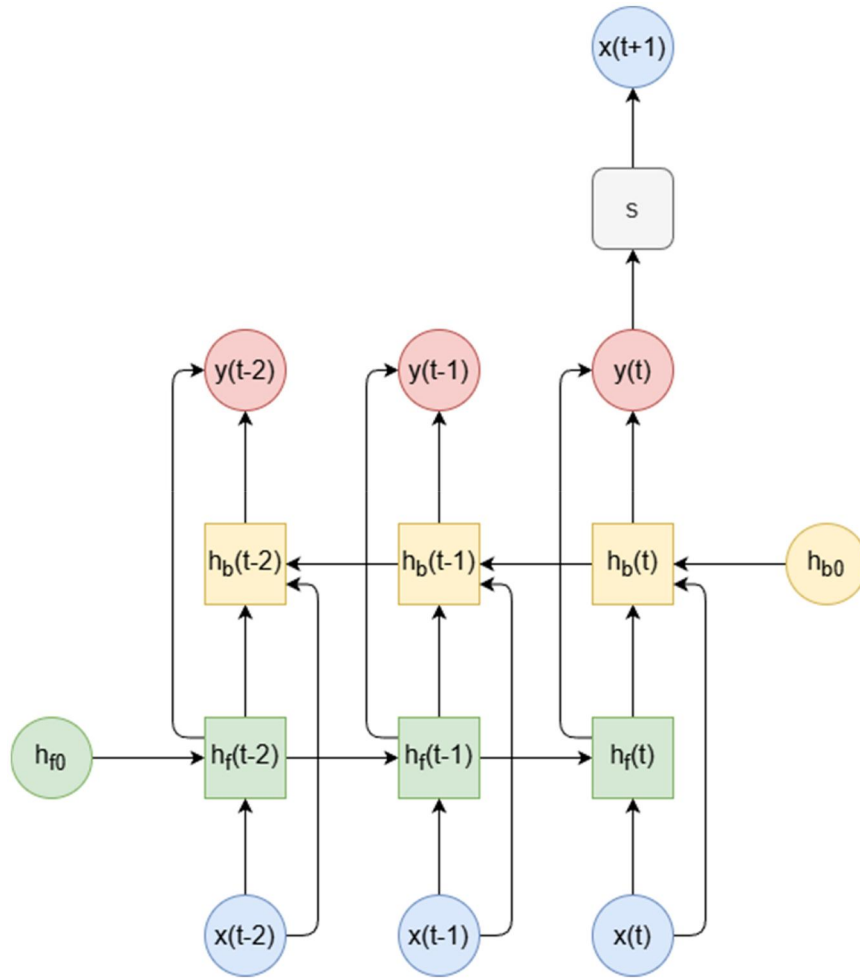


FIG. 1. Diagram of the neural network.

The calculations in the neural network model are conducted in the following order:

- (1) Converting the input value $x(t)$ to the hidden state $h_f(t)$ of the forward sequence, while taking into account the previous hidden state $h_f(t-1)$,

$$h_f(t) = f_1(U_1 x(t) + U_2 h_f(t-1) + b). \quad (2)$$

- (2) Calculating the hidden state $h_b(t)$ of the reverse sequence, while taking into account the subsequent hidden state $h_b(t+1)$, the current hidden state $h_f(t)$, and the input value $x(t)$,

$$h_b(t) = f_2(V_1 x(t) + V_2 h_f(t) + V_3 h_b(t+1) + c). \quad (3)$$

- (3) Transforming the hidden states $h_f(t)$, $h_b(t)$ into the output of the recurrent network,

$$y(t) = f_3(W[h_f(t), h_b(t)] + d), \quad (4)$$

where the square brackets $[,]$ denote the concatenation operation.

- (4) Converting the output of the recurrent network to the dimension of the initial configuration using a fully connected layer and obtaining a prediction of the configuration of atoms $x(t+1)$ at the next time step,

$$x(t+1) = s(Ry(t) + q). \quad (5)$$

In Eqs. (2)–(5), the terms $U_{1,2}$, $V_{1,2,3}$, W , and R are the weight matrices, and b , c , d , and q are biases that are to be determined during the training of a neural network. The neural network is implemented using the PyTorch package.³⁴ The source code files implementing the training and inference of the network are available in GitHub repository.³⁵

The results from simulating copper bulk in LAMMPS package³⁶ with EAM potential³⁷ were used as a training sample. In general, neural networks in classical molecular dynamics are

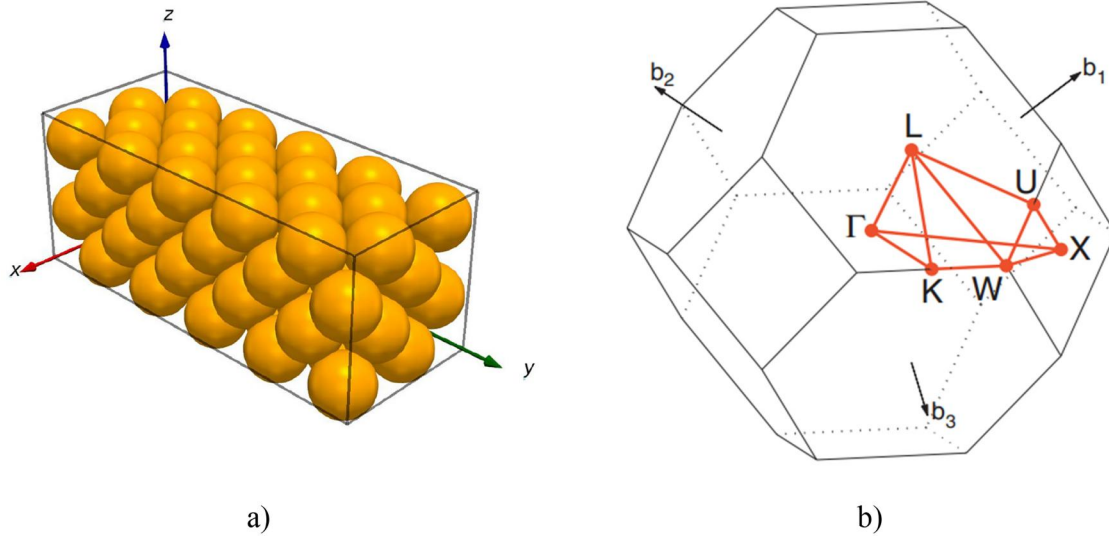


FIG. 2. Face-centered cubic copper lattice: (a) arrangement of atoms; (b) arrangement of points of high symmetry in the reverse space for a face-centered cubic lattice. (b) Reproduced with permission from Setyawan and Curtarolo, *Comput. Mater. Sci.* **49**(2), 299–312 (2010). Copyright 2010 Elsevier.⁴³

trained on *ab initio* calculation data, since the main point is to take into account unknown quantum mechanical patterns in particle interactions in the force field. This article aims to demonstrate the idea of using a recurrent neural network with a simple architecture to predict some statistical properties of solids. Therefore, in order to reduce the time of experiments, training was conducted on data from classical MD.

The modeling area is represented by a rectangular parallelepiped of $2 \times 5 \times 2$ copper unit cells [see Fig. 2(a)] with periodic boundary conditions that contain 80 atoms [thus, each configuration of type (1) contains 240 real numbers]. The reason for extending the length of the sample in the direction of the OY axis in comparison with the other directions is that the dynamic structural factor, which is a figure of merit for evaluating the states of the crystal under consideration predicted by the neural network, will be calculated along this direction. After minimizing the energy, the atomic velocities were randomized with the Maxwell distribution at temperature of 1 K. After that, the temperature increased evenly up to 300 K over 200 ps in NVE ensemble, and for another 200 ps, the system was maintained at 300 K in NVT ensemble with the Berendsen thermostat. After such preparation of the model system, a sequence of frames of particle positions, used for training the neural network, was recorded. The time step was set to 2 fs during the entire simulation.

The training procedure is conventional for this type of network: a block of consecutive frames of a given length was randomly selected in the array of trajectory frames and divided into consecutive triples of frames. A training sample is considered as a certain number of these triples, which are presented in random order during training. The loss function is L2.

Next, using the optuna library,³⁸ a set of optimal hyperparameters was determined: the size of the training batch, the learning

rate, the size of the training sample, and the size of the weight matrices of the neural network. The dynamic structural factor along the direction of the OY axis was used as the main comparative value used in the loss function when optimizing hyperparameters and later, when selecting models from a number of trained ones. The dynamic structural factor is one of the characteristics that determines the mutual correlation of the states of the system under consideration. We will provide the necessary information about and definition of the dynamic structural factor.^{1,39,40}

The equilibrium correlation function over time of two dynamic variables $A(t)$ and $B(t)$ is determined by the expression³⁹

$$C_{AB}(t', t'') = \langle A(t')B^*(t'') \rangle, \quad (6)$$

where $t'' \geq t'$ and the angle brackets express time averaging

$$\langle A(t')B^*(t'') \rangle = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A(t' + t)B^*(t'' + t)dt, \quad (7)$$

which, in view of the ergodic hypothesis, can also be interpreted as an ensemble averaging.

The spatial Fourier image of the correlation function is used to calculate the dynamic structural factor,

$$C_{AB}(q; t', t'') = \langle A_q(t')B_q^*(t'') \rangle = \langle A_q(t')B_{-q}(t'') \rangle. \quad (8)$$

The particle concentration is given by the expression³⁹

$$n(\mathbf{r}, t) = \sum_i^N \delta(\mathbf{r} - \mathbf{r}_i(t)), \quad (9)$$

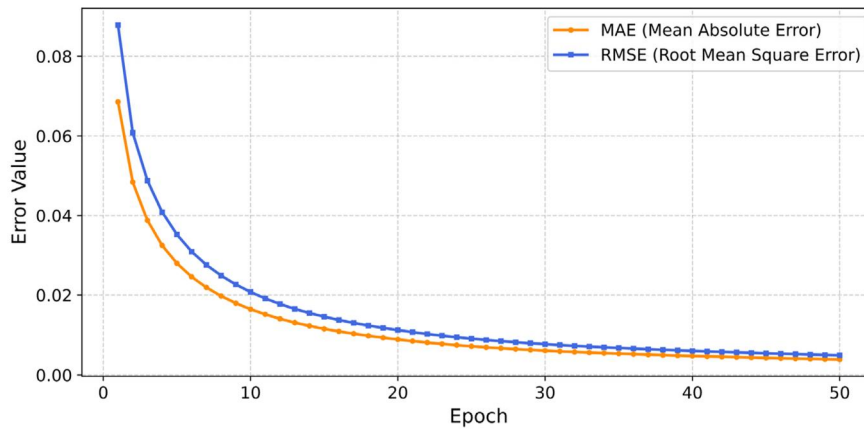


FIG. 3. Learning curves of RNN networks.

where $\mathbf{r}_i(t)$ is the radius vector that defines the position of the i th particle at a given time t and $\delta(x)$ is the Dirac delta function. The correlation function of the spatial Fourier components of the concentration is represented by the following expression:

$$F(\mathbf{q}, t) = \frac{1}{N} \langle n(\mathbf{q}, t) n(-\mathbf{q}, 0) \rangle = \frac{1}{N} \sum_i^N \sum_j^N \langle \exp[i\mathbf{q} \cdot (\mathbf{r}_i(t) - \mathbf{r}_j(0))] \rangle, \quad (10)$$

where N is the number of particles in the system. The dynamic structural factor is derived through the time Fourier image of (10),

$$S(\mathbf{q}, \omega) = \int_{-\infty}^{\infty} F(\mathbf{q}, t) \exp(-i\omega t) dt. \quad (11)$$

The dynasor package⁴¹ is used in this work to calculate the dynamic structural factor based on MD data.

The following procedure was performed to compute the value of the loss function:

- (1) The sequence of three frames was randomly selected from the array of frames that did not participate in the training and was specified as the initial state of the model.
- (2) The model then generated the following atomic configurations over a given time (ranged from 20 to 100 ps), and the dynamic structural factor $S(\mathbf{q}, \omega)$, normalized to the maximum value, was calculated on the frames, obtained during the simulation.
- (3) Similarly, the dynamic structural factor $S_0(\mathbf{q}, \omega)$ was calculated on the same number of frames from the target array that follows after the selected three initial frames.
- (4) Next, the Euclidean norm of the difference $S(\mathbf{q}, \omega) - S_0(\mathbf{q}, \omega)$ was calculated, which served as the value of the loss function.

This function was used both in hyperparameter optimization and further in model selection, with the only difference being that the average value for several simulations was used when selecting the model.

IV. RESULTS AND DISCUSSIONS

Figure 3 shows the learning curves of an RNN network on train sample—graphs of mean absolute error (MAE) and root mean square error (RMSE) on the epoch number. There is a steady decrease in the values of both metrics, which indicates that the training is correct.

On the other hand, the learning curves themselves are only an indicator that the neural network “assimilates” the information presented to it, but they do not provide estimation of the quality of the representation of the ensemble of particles by the neural network under consideration. An objective analysis of the results obtained is based on consideration of quantities that take into account the statistical properties of the particle distribution—the dynamic structural factor $S(\mathbf{q}, \omega)$, the radial distribution function (RDF), and the velocity autocorrelation function (VACF).

Figure 4 shows typical graphs of $S(\mathbf{q}, \omega)$ and $S_0(\mathbf{q}, \omega)$ for one of the adequate models selected according to the described methodology. Since a crystal fragment under consideration includes five elementary copper cells located along the OY axis, the graphs of $S(\mathbf{q}, \omega)$ and $S_0(\mathbf{q}, \omega)$ are constructed depending on the component of the wave vector q_y , which takes, respectively, five discrete values. A cautious conclusion can be made from the graphs, that the trajectories, predicted by a simple RNN model, inherit the statistical properties of the initial system.

Figure 5 shows a quantitative analysis of the degree of coincidence between the graphs of the dynamic structural factor, constructed from the data of the initial molecular trajectories $S_0(\mathbf{q}, \omega)$ [Fig. 5(a)] and from the predictions of the RNN network $S(\mathbf{q}, \omega)$ [Fig. 5(b)].

In Figure 5(c), it is presented a heat map of the absolute value of difference between $S(\mathbf{q}, \omega)$ and $S_0(\mathbf{q}, \omega)$; in Fig. 5(d), it is presented a graph of the correlation coefficient between the values of the raw data of dynamic structural factors $S_0(\mathbf{q}, \omega)$ and $S(\mathbf{q}, \omega)$, calculated for corresponding values of q_y .

The Pearson correlation coefficient between the vectors $x = \{x_1, x_2, \dots, x_N\}$ and $y = \{y_1, y_2, \dots, y_N\}$ is determined by

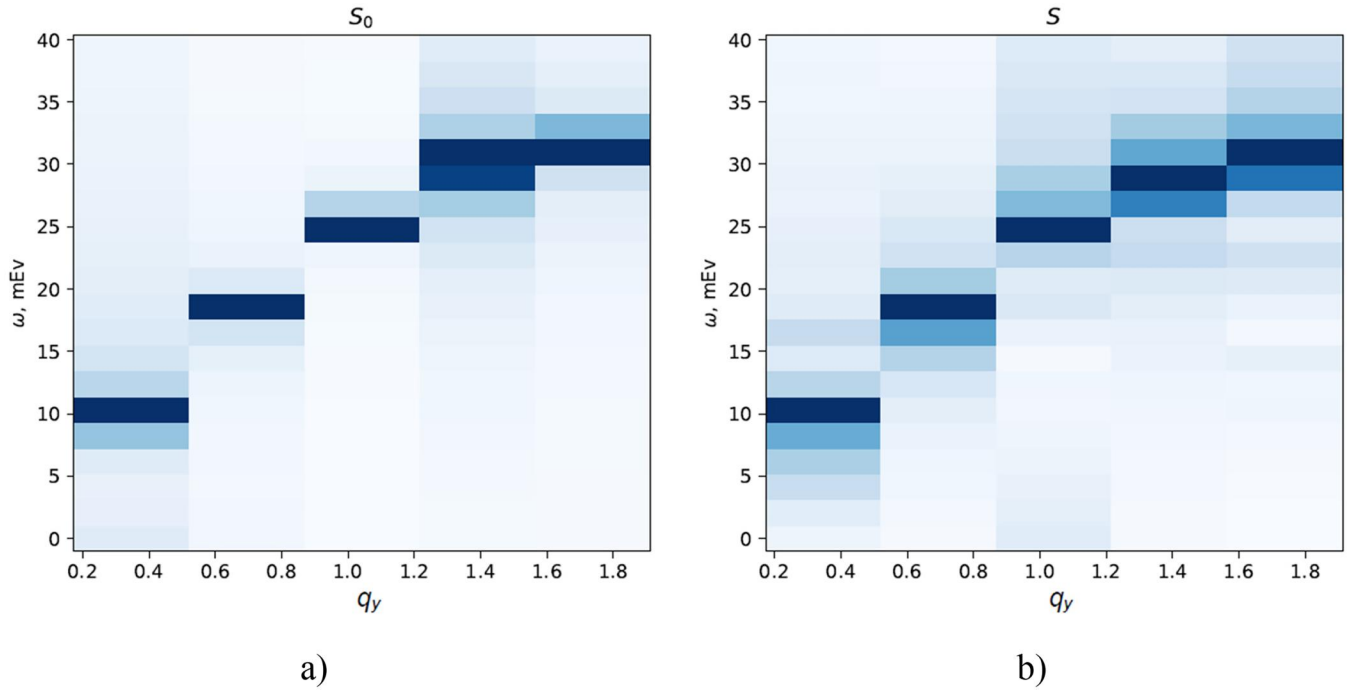


FIG. 4. Values of the dynamic structural factor $S_0(\mathbf{q}, \omega)$ calculated from the molecular trajectories from the training sample (a) and the dynamic structural factor $S(\mathbf{q}, \omega)$ calculated from the molecular trajectories predicted by the RNN network (b). The component q_y of quasi-wave vector takes values on the path between the points Γ and X in the reciprocal space (see Fig. 2).

the following expression:

$$r_{xy} = \frac{\sum_j (x_j - \bar{x})(y_j - \bar{y})}{\sqrt{\sum_j (x_j - \bar{x})^2 \sum_j (y_j - \bar{y})^2}}, \quad (12)$$

where $\bar{x} = \frac{1}{N} \sum_{j=1}^N x_j$ and $\bar{y} = \frac{1}{N} \sum_{j=1}^N y_j$ are mean values. In Fig. 5(d), it is presented the dependence of correlation coefficient $r_{SS_0}(q_y)$ between the values of dynamic structural factor $S_0(q_y^i, \omega)$ and $S(q_y^i, \omega)$, or, in other words, between corresponding columns of heat maps, plotted in Fig. 5(a) and Fig. 5(b), on wave vector component q_y ,

$$r_{SS_0}(q_y^i) = \frac{\sum_j (S_0(q_y^i, \omega_j) - \bar{S}_0(q_y^i, \omega)) (S(q_y^i, \omega_j) - \bar{S}(q_y^i, \omega))}{\sqrt{\sum_j (S_0(q_y^i, \omega_j) - \bar{S}_0(q_y^i, \omega))^2 \sum_j (S(q_y^i, \omega_j) - \bar{S}(q_y^i, \omega))^2}}. \quad (13)$$

From the graphs in Figs. 5(c) and 5(d), one can see that the difference between dynamic structural factor values, calculated

elementwise for the matrices $S_0(q_y^i, \omega_j)$ and $S(q_y^i, \omega_j)$, is great enough, and a minimal value of the correlation coefficient $r_{SS_0}(q_y^i)$ is equal to 0.46, which is the evidence of weak correlation. On the other hand, maximum values on the heat map of $S(\mathbf{q}, \omega)$ determine the dispersion curve of one of the acoustic phonon modes in the considered material. The shape of this curve is clearly visible in Figs. 4(a) and 4(b) and in Figs. 5(a) and 5(b). It is more convenient to trace statistically the degree of coincidence of the dispersion curves corresponding to $S_0(q_y^i, \omega_j)$ and $S(q_y^i, \omega_j)$ if we introduce Gaussian blur of the heat maps on the values of the cyclic frequency ω . Figure 5(e) shows a heat map of the absolute value of the difference in the values of dynamic structural factors $|S(\mathbf{q}, \omega) - S_0(\mathbf{q}, \omega)|$, Fig. 5(f) shows the dependence of the correlation coefficient $r_{SS_0}(q_y)$ calculated from the data $S_0(\mathbf{q}, \omega)$, $S(\mathbf{q}, \omega)$, to the heat maps of which the Gaussian blur operation was applied. There is a significantly smaller difference between $S_0(\mathbf{q}, \omega)$ and $S(\mathbf{q}, \omega)$ and a significantly higher correlation coefficient between the columns of the heat maps corresponding to the same values of q_y compared to the graphs shown in Figs. 5(c) and 5(d).

Gaussian blur was performed using a convolution operation,

$$\tilde{S}(q_y, \omega) = \sum_i S(q_y, \omega_i) G(\omega - \omega_i), \quad (14)$$

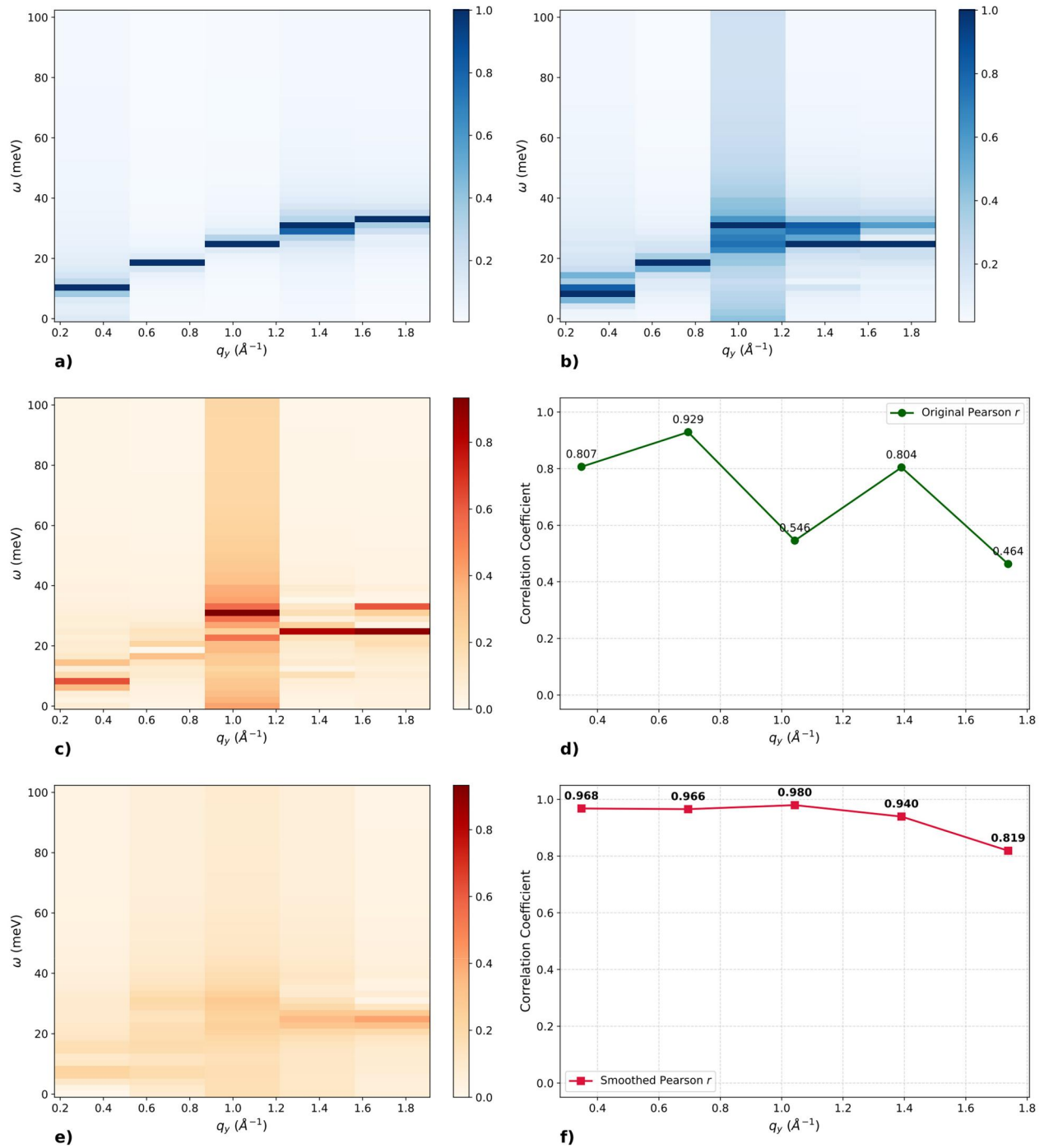


FIG. 5. Comparison of heat maps of the dynamic structural factor; (a) values of the dynamic structural factor $S_0(\mathbf{q}, \omega)$ calculated from the molecular trajectories based on the training sample; (b) values of dynamic structural factor $S(\mathbf{q}, \omega)$ calculated from the molecular trajectories predicted by the RNN network; (c) absolute values of difference $|S(\mathbf{q}, \omega) - S_0(\mathbf{q}, \omega)|$, calculated by original data of the dynamic structural factors, presented on (a) and (b) plots; (d) graph of the correlation coefficient between the values of the raw data of dynamic structural factors $S_0(\mathbf{q}, \omega)$ and $S(\mathbf{q}, \omega)$, calculated for fixed values of q_y ; (e) absolute values of difference $|S(\mathbf{q}, \omega) - S_0(\mathbf{q}, \omega)|$, calculated by the dynamic structural factors, which are smeared by Gaussian blur on frequency ω at fixed q_y values; (f) graph of the correlation coefficient between the dynamic structural factors $S_0(\mathbf{q}, \omega)$ and $S(\mathbf{q}, \omega)$, which are smeared by Gaussian blur on frequency ω at fixed q_y values.

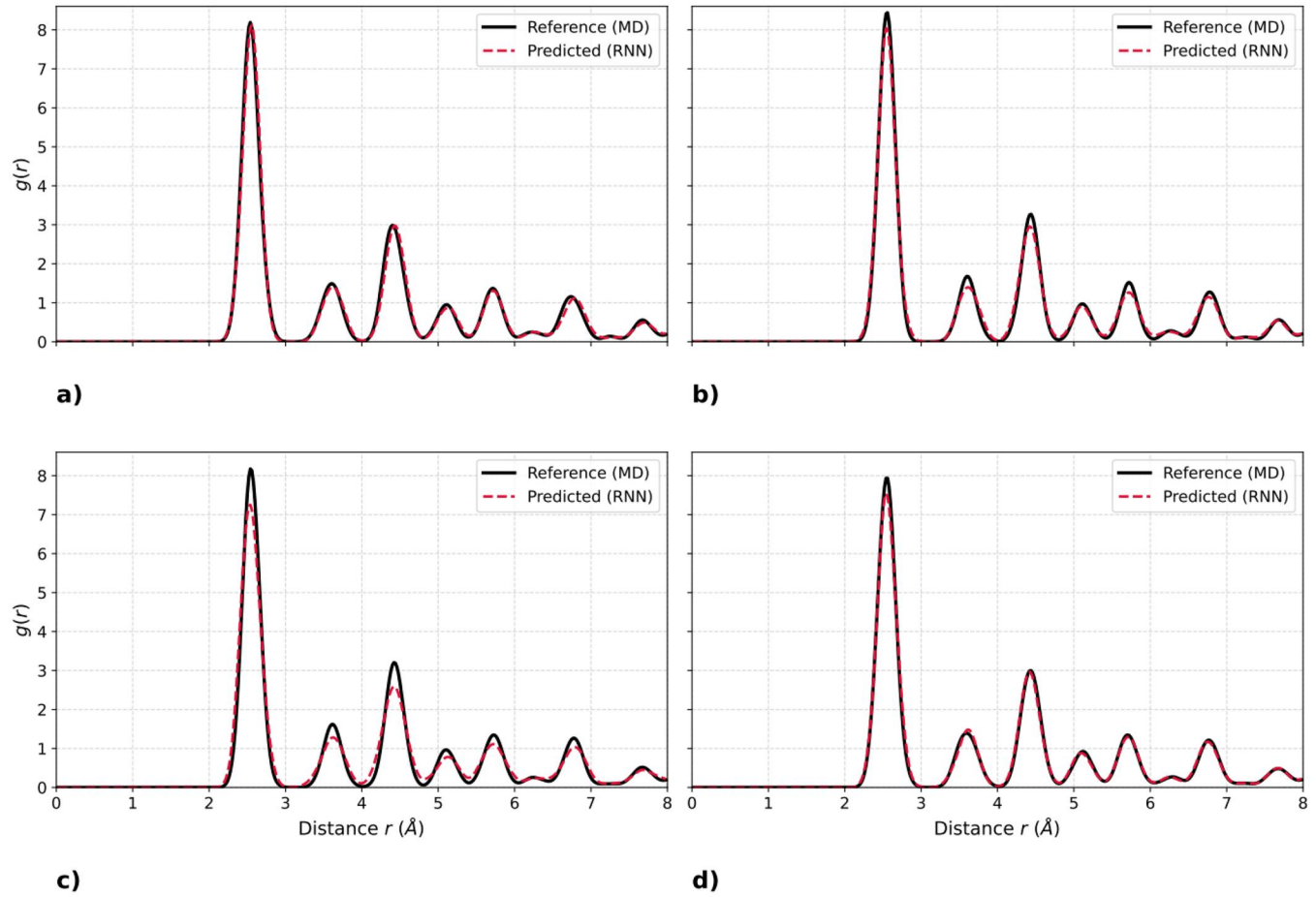


FIG. 6. Graphs of the radial distribution function $g(r)$ at specific time frames: (a) 20, (b) 100, (c) 140, (d) 180 ps from the beginning of the following up on the system's state.

$$G(\omega) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\omega^2}{2\sigma^2}\right), \quad (15)$$

where σ determines the blur radius. During the calculations it was assumed $\sigma = 1.0$.

Figure 6 shows graphs of the radial distribution function, and Fig. 7 shows histograms of first-order coordinate differences (which are up to a factor equal to particle velocities) obtained as a result of simulation over 180 ps. The initial three frames of atomic configurations are the same for the simulation using the RNN and for the control sample obtained during direct MD simulation. Figures 6 and 7 show that the network retains the statistics of the coordinates and velocities of the test sample well during the simulation, and there is no noticeable degradation of these dependencies.

Let us consider the velocity autocorrelation function, which shows how the velocity of a particle at a given time is related to its

velocity at a previous moment. The normalized velocity autocorrelation function is defined by the expression

$$VACF(\tau) = \frac{\langle \mathbf{v}(t_0) \mathbf{v}(t_0 + \tau) \rangle}{\langle |\mathbf{v}(t_0)|^2 \rangle}, \quad (16)$$

where the angle brackets define the averaging of the ensemble of particles and all possible initial time points t_0 ,

$$VACF(\tau) = \frac{\frac{1}{T-\tau} \sum_{t_0=0}^{T-\tau-1} \left(\sum_{i=1}^N \mathbf{v}_i(t_0) \mathbf{v}_i(t_0 + \tau) \right)}{\frac{1}{T} \sum_{t_0=0}^{T-1} \left(\sum_{i=1}^N |\mathbf{v}_i(t_0)|^2 \right)}. \quad (17)$$

Here, t_0 is the beginning of the time interval with duration τ , T is the length of the time interval corresponding to the complete

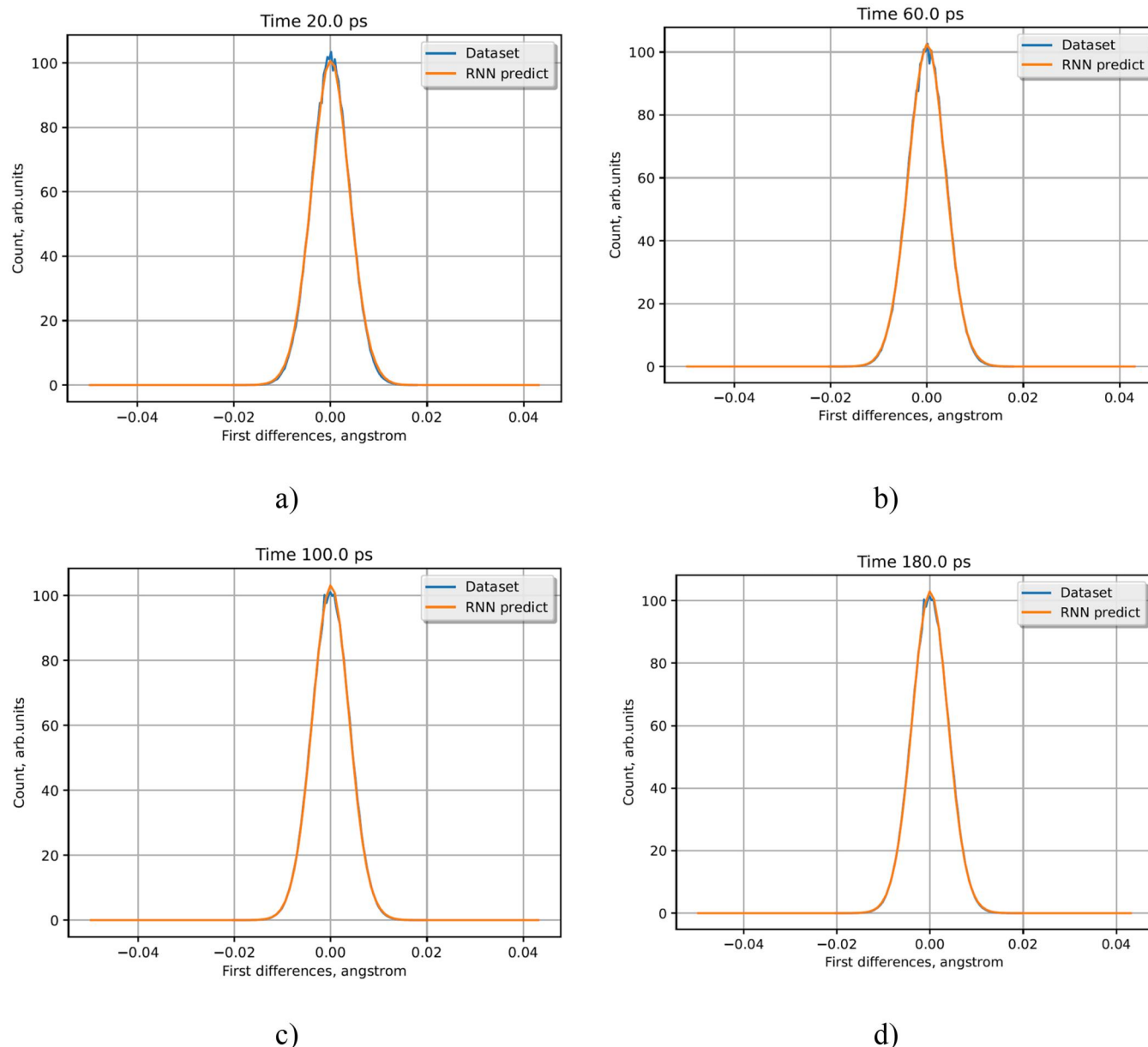


FIG. 7. Histograms of the first coordinate difference at time of 20 (a), 60 (b), 100 (c), 180 ps (d) from the beginning of the following up on the system's state.

molecular trajectory, $\mathbf{v}_i(t)$ is the velocity vector of the i th atom at time t , and N is the total number of particles. The values t_0 , τ , and T take discrete values with step Δt determined by the initial data. The calculations assumed $\Delta t = 2$ fs, $T = 100\,000$, τ takes values from 0 to 100, $N = 80$. Figure 8(a) shows VACF graphs based on the reference molecular dynamics data and data predicted by the RNN network. There is a good match between the VACF graphs.

Figure 8(b) shows a graph of velocity absolute deviation (VAD), defined as the integral of the modulus of the difference between the two specified VASF curves over a time interval $[0, \tau]$. The VAD value allows you to quantify the divergence of VACF values plotted on the initial data, and the data predicted by the RNN network.

Let us consider how much the heat map of the dynamic structural factor $S(\mathbf{q}, \omega)$, calculated by means of the RNN

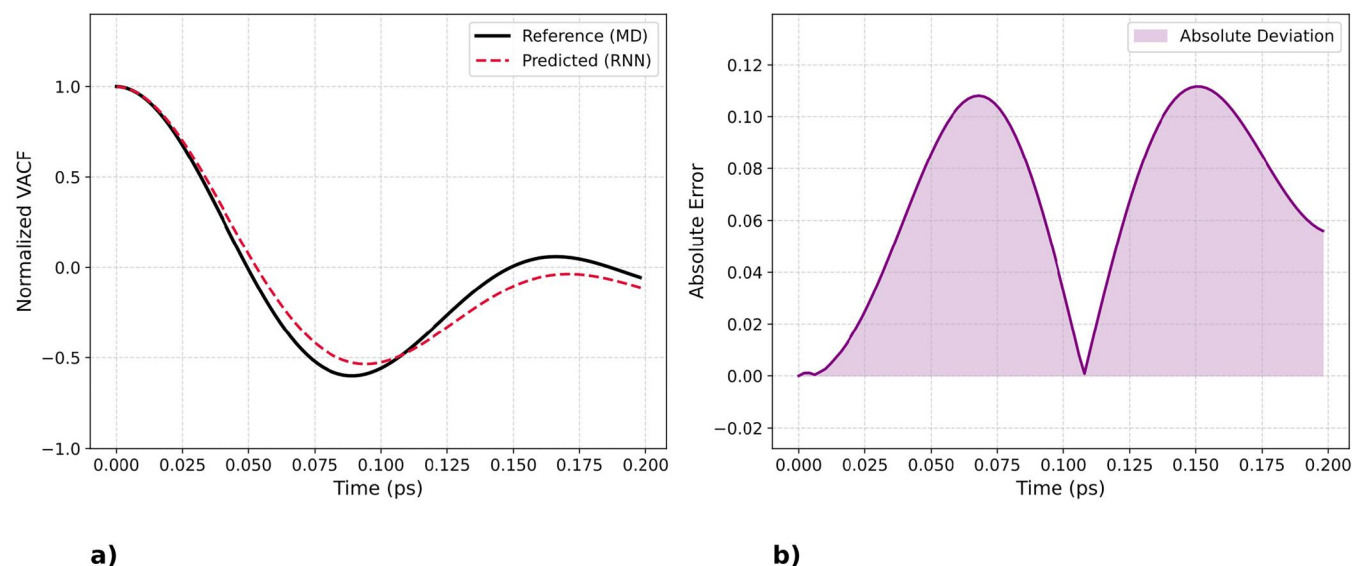


FIG. 8. Velocity autocorrelation function (a) and velocity absolute deviation (b).

network, changes at different points in time, compared with the initial values. The molecular dynamics data used for training is calculated for a range of 200 ps. To evaluate the quality of the representation of the statistical characteristics of the particle distribution by the developed neural network, it is necessary to take the length of the time interval longer than the duration of the initial molecular trajectories. We took the length of the time interval to be 500 ps (250 000 time steps of 2 fs). The dynamic structural factor can be calculated for a certain time interval. In the work, the length of the interval was assumed to be equal to 2000 time steps, a total of six values of such intervals were taken. The point on the graph (Fig. 9) corresponds to the center of the interval. Figure 9 shows graphs of the correlation coefficient between heat maps $S_{t=0}(q_y, \omega)$ and $S_{t=t_i}(q_y, \omega)$, calculated based on the exact values and on the values obtained after applying the Gaussian blur procedure. As noted above, calculating the correlation coefficient from smoothed data makes it possible to reduce the influence of local fluctuations and assess the overall picture of the statistical distribution. It can be seen from the graphs in Fig. 9 that the correlation coefficient, based on the values $S(q_y, \omega)$ without smoothing, takes values from 0.461 to 0.759, while using smoothed data, the correlation coefficient values range from 0.737 to 0.886. Figure 10 shows a graph of the correlation coefficient of the radial distribution function $g(r)$ at the current time with the initial values, plotted on a time interval of 500 ps. It can be seen that the correlation coefficient retains a value very close to 1.0. Based on the obtained dependence of the correlation coefficient between the initial heat map of $S_{t=0}(q_y, \omega)$ and the heat maps of $S_{t=t_i}(q_y, \omega)$ at the following time points, as well as the dependence of the correlation coefficient of the radial distribution function, it can be concluded that the statistical properties of the particle ensemble of the RNN network studied in this work are preserved over a sufficiently long time interval.

Below are the main advantages and disadvantages of the described approach for calculating statistical characteristics. The disadvantages include, first of all, that this technique in its current form works only for crystals and, in addition, the trained model is suitable only for working with the exact configuration of the supercell on which it is trained and with the same time step. However, these limitations do not seem insurmountable, and they are the subject of further research.

The main advantage of this approach is, first of all, the speed of operation—both the choice of the most accurate model for atomic configuration prediction and its inference are much faster than classical MD calculation of those using force representation models based on deep neural networks (DeePMD and PaiNN) trained by data of quantum chemical calculations.

It should be noted that in Ref. 20, the models based on LSTM (long short-term memory) and GRUs (gated recurrent units) were used, whose architectures are designed to predict the next configuration based on a rather long chain of previous states. At the same time, the calculation of molecular trajectories, which are determined by Newton's equations of motion, requires knowledge of the configurations of the system only in the two previous time steps.

In this work, a recurrent neural network of the simplest form was suggested, which has three hidden layers, corresponding to taking into account the two previous states to calculate the next one. Calculations have shown that the model, constructed with the described methodology, is not inferior in accuracy to LSTM nor GRU networks, and as an advantage, it requires significantly shorter training and initial sequences (three instead of 50–200 in Ref. 20), small training sample size and, accordingly, less time for training. With the same number of hidden layers, the RNN network of the architecture under consideration shows a training

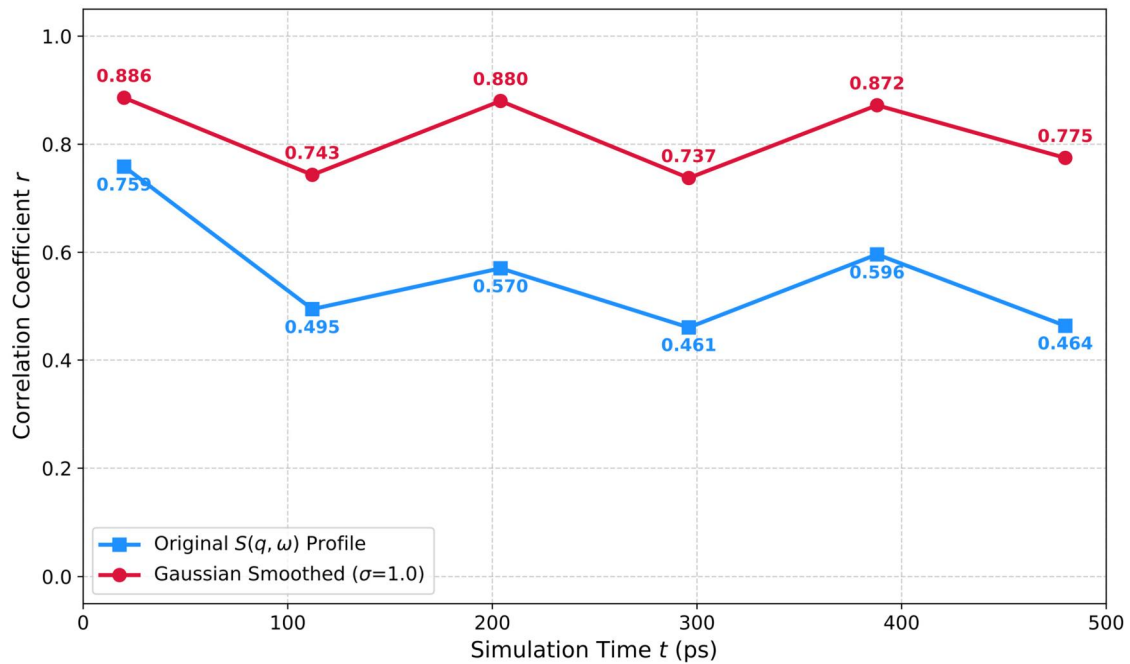


FIG. 9. Dynamic structural factor $S(q, \omega)$ correlation with the initial state.

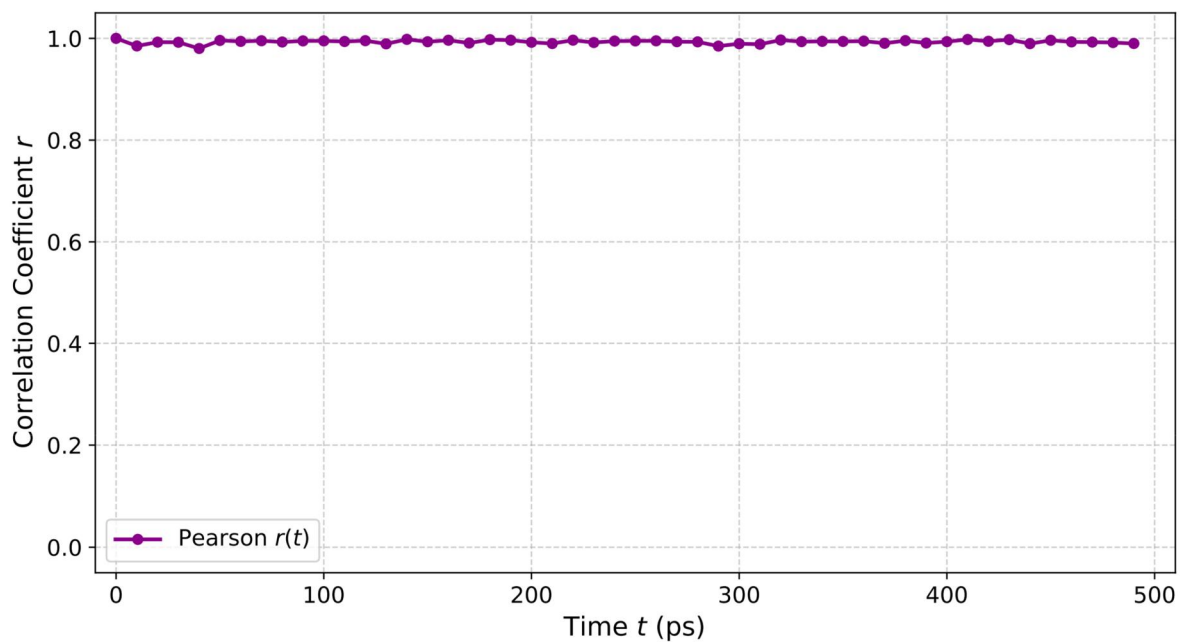


FIG. 10. Radial distribution function $g(r)$ correlation with the initial state.

and inference time between half and a quarter as much than that of the LSTM or GRU architecture networks.

The small size of the training set sufficient for the recurrent neural network presented in this paper allows us to propose its following application. Suppose a set of molecular trajectories is obtained for a sample of some crystalline material using the method of quantum molecular dynamics. For example, starting from a certain point in time, considered molecular trajectories should correspond to thermodynamic equilibrium. The microstates of the thermodynamic system under consideration obtained on the basis of these molecular trajectories will account for only a small fraction of the total number of microstates corresponding to the macrostate under study. Therefore, when using AIMD data as a training set for a neural network that calculates the potential of interatomic interaction (actually, force field in the considered material),^{5,12} it is a possible case (see, for example, Ref. 42) when the neural network does not approximate the force field well due to the lack of a sufficient number of statistically independent states in the training set. The recurrent network proposed in this paper, which preserves the statistical properties of the distribution, can generate a large number of spatial configurations of atoms for which forces can be calculated and the corresponding values can be included in the training set to predict the interatomic potential.

V. CONCLUSIONS

Thus, the paper substantiates the use of bidirectional recurrent neural networks with a minimum number of hidden layers to predict the results of molecular dynamics using the example of modeling the state of thermodynamic equilibrium of a copper crystal. A network that takes into account the states (atomic configurations) of the system at the current and two previous time steps to predict the state of the system at the next step is considered. Based on a comparison of heat maps of the dynamic structural factor, histograms of the first differences in particle coordinates (in fact, up to a multiplier of particle velocities) as well as graphs of the radial distribution function, which takes into account the distribution of relative distances between particles, it can be concluded that the positions of particles predicted by the neural network retain the properties of a statistical distribution corresponding to direct MD modeling data. The aim of further research is to determine whether the neural network under consideration can adequately reproduce other correlators, primarily particle velocity correlators, through which phonon frequencies can be expressed.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

D. V. Zav'yalov: Conceptualization (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal). **V. I. Konchenkov:** Investigation (equal); Writing – original draft (equal). **E. S. Sivashova:** Data curation (equal); Visualization (equal). **D. V. Shein:** Investigation (equal); Software (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in GitHub at https://github.com/sinegordon/crystal_rnn, Ref. 35.

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